Division-polynomial covariance and Kummer sequences on secp256k1

Exact character targets, disjoint branch layers, and limits of finite-orbit inference

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Document classification: Public AI-developed mathematical research artifact – independently unverified candidate results

Status and reading guide. This document records a human-initiated, substantially AI-developed mathematical investigation. The human curator directed the objectives but does not independently certify its advanced mathematical arguments. The novel claims and supplied proofs are candidate results awaiting independent specialist verification; public availability does not certify correctness, completeness or originality. This artifact proposes Kummer covariance obstructions, recurrence and word-occurrence bounds, and higher-genus quotient/isogeny constructions. The integer threshold checks do not verify the underlying character-sum estimates, geometric independence, conductor assumptions or isogeny/descent arguments. Those steps require independent mathematical examination. No unrestricted private-key-recovery claim is made.

Abstract

The division-polynomial investigation is organized around an exact covariance equation: a base-field sextic character of a rational function is sought that reproduces the sextic character of a hidden scalar. On secp256k1, a primitive degree-three CM endomorphism turns the question into a cyclic-character problem. The manuscript proposes a divisor calculation for a division-polynomial covariance ratio, identifies a cubic atom, and investigates an obstruction to the separable single-division-polynomial family as exact scalar-character extractors. A proposed conductor argument yields large effective divisor support for products. Along the primitive orbit, the cubic atom defines a ternary sequence with nontrivial holonomy; proposed disjoint branch layers of sizes $18 \cdot 3^i$ lead to recurrence and block-occurrence estimates. Subject to the stated geometric and character-sum hypotheses, the manuscript proposes exclusion of nonzero affine recurrences of length at most 78 and occurrence of every ternary word of length at most 39. It also develops a cubic-extension Kummer-symbol model, genus-seven covering geometry, candidate quotient curves, a Jacobian decomposition, and a degree-three isogeny between elliptic quotients of a reciprocal genus-two model. Independence and descent assumptions are separated explicitly. These proposals concern restricted observables, not

general private-key recovery, and require specialist examination.

Keywords: elliptic divisibility sequence; division polynomial; Kummer cover; cubic character; CM endomorphism; cyclic sequence; secp256k1.

1. Setting and conventions

The relationship between elliptic divisibility sequences and discrete logarithms is established prior work [4]. Here the question is narrower: can a rational function's sextic character reproduce the multiplicative sextic character of an unknown scalar, at every nonzero point of one fixed prime-order group? The exploratory investigation used numerical evaluations to test the initial intuition; the exact covariance equation below replaces any statistical criterion in the mathematical argument.

Use the SEC 2 curve [5]

$$E/\mathbf{F}_p : y^2 = x^3 + 7, \qquad p = 2^{256} - 2^{32} - 977.$$

The group order n and the two oriented cube roots are specified by concatenating each pair of hexadecimal lines:

```
n      = FFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFE
         BAAEDCE6AF48A03BBFD25E8CD0364141
beta   = 7ae96a2b657c07106e64479eac3434e9
         9cf0497512f58995c1396c28719501ee
lambda = 5363ad4cc05c30e0a5261c028812645a
         122e22ea20816678df02967c1b23bd72
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SEC 2 specifies a generator G of this prime-order group. The identity $\phi(G) = [\lambda]G$, where $\phi(x, y) = (\beta x, y)$, is checked by the ordinary group law on its published coordinates. Hence $\phi = [\lambda]$ on all rational points. We fix this orientation, rather than interchanging the two cube roots implicitly.

Put $H = \{\pm 1, \pm\lambda, \pm\lambda^2\}$, $h = 2 + \lambda$, $\Psi = 2 + \phi = 1 - \phi^2$, and $M = (n-1)/6$. Appendix A prints the complete certificate that h has order $n - 1$. The map Ψ is separable of degree three: its kernel is $\{O, (0, \sqrt{7}), (0, -\sqrt{7})\}$, and its differential is nonzero. Alternatively its Eisenstein norm is three. The chord from P to $-\phi^2 P$ gives

$$x(\Psi P) = -\frac{\beta\, x^3 + 28}{3 \quad x^2}, \qquad F(t) = -\frac{(t+28)^3}{27t^2}, \quad t = x^3. \tag{1.1}$$

Indeed the slope is $-2y/[x(\beta^2 - 1)]$, and $(\beta^2 - 1)^2 = -3\beta^2$. Cubing the abscissa gives $t(\Psi P) = F(t(P))$. The identity $\Psi^2 = -[3]\phi^2$ follows by expanding $(2 + \phi)^2$ and using $1 + \phi + \phi^2 = 0$.

The coordinate t is a complete sixfold quotient: equality of two cubes of abscissae gives $x(Q) = \beta^j x(P)$, and the curve equation then gives $y(Q) = \pm y(P)$. No nonidentity rational point has a nontrivial stabilizer, since that would give two- or three-torsion. Primitivity of h therefore makes Ψ a single $(n-1)$-cycle on nonzero points and F a single M-cycle on their quotient.

For nonzero field elements set

$$\eta(z) = z^{(p-1)/6}, \qquad \chi_3(z) = z^{(p-1)/3} = \eta(z)^2.$$

The values here are initially roots of unity in $\mathbf{F}_p$. Whenever absolute values or character sums are used, identify β with $e^{2\pi i/3}$ and -1 with the usual complex number -1. This specifies a compatible complex character. Let $\iota : \mu_6(\mathbf{F}_n) \to \mu_6(\mathbf{F}_p)$ send λ to β and -1 to -1. Appendix A verifies

$$h^M = -\lambda^2, \quad M \equiv 1 \pmod 3, \quad p \equiv 7 \pmod 9, \quad \chi_3(\beta/81) = 1. \qquad (1.2)$$

All multiplication indices below are positive integers prime to p; indices divisible by n are also excluded when evaluating a division polynomial on every nonidentity rational point.

2. The exact sextic target and covariance

Let $P_d = [d]G$ for $d \in \mathbf{F}_n^*$ and define the scalar label

$$C(P_d) = \iota(d^M).$$

The proposed exact extractor is a rational function $g \in \mathbf{F}_p(E)^*$, finite and nonzero at every P_d, with

$$\eta(g(P_d)) = C(P_d). \qquad (2.1)$$

Unlike its covariance ratio, this label is not invariant under all sixfold scalar symmetries. With $\xi = \iota(h^M) = -\beta^2$, it satisfies $C(\Psi P) = \xi C(P)$.

Proposition 2.1 (orbit covariance). Put $R_g = (g \circ \Psi)/g$. A necessary condition for (2.1) is

$$\eta(R_g(P)) = \xi. \qquad (2.2)$$

Equation (2.2) is required at every nonidentity rational point. Conversely, (2.2) together with (2.1) at one point implies (2.1) everywhere on that set.

Proof. Taking the ratio of (2.1) at ΨP and P gives (2.2). Conversely, propagation along the single $(n-1)$-cycle of Ψ determines the label from one anchor. □

Thus the exact target has a concrete mathematical meaning. A weak correlation on sampled points, or an identity valid only after knowing d, is not a substitute for (2.2).

3. Division-polynomial covariance divisors

Let ψ_m be the standard division polynomial. Its divisor, for $p \nmid m$, is [1]

$$\operatorname{div}(\psi_m) = [m]^*(O) - m^2(O). \qquad (3.1)$$

Over an algebraic closure, write $\tau = (0, \sqrt{7})$. The kernel of Ψ is $\{O, \tau, -\tau\}$, and τ has order three. Define

$$R_m(P) = \frac{\psi_m(\Psi P)}{\psi_m(P)}.$$

Proposition 3.1. The covariance divisor is

$$\operatorname{div}(R_m) = [m]^*((\tau) + (-\tau)) - m^2((\tau) + (-\tau)). \qquad (3.2)$$

Moreover R_m is invariant under negation and ϕ, so it belongs to $\mathbf{F}_p(t)$.

Proof. Pull (3.1) back by Ψ and subtract (3.1). Since the two endomorphisms commute and $\Psi^*(O) = (O) + (\tau) + (-\tau)$, the terms involving (O) cancel and give (3.2). The division-polynomial transformation factors under negation and the cubic automorphism are constants; those constants cancel in the ratio. Thus R_m descends through the sixfold quotient. □

If $n \nmid m$, neither P nor ΨP is killed by $[m]$ for a nonidentity rational P. Hence R_m is finite and nonzero at every such point. The standard composition identity gives the useful exact relation

$$R_{ab}(P) = R_a([b]P) R_b(P)^{a^2}. \qquad (3.3)$$

This formula reduces character behavior of composite indices to atomic factors evaluated at known multiples. It does not reveal the unknown scalar of P.

4. The cubic atom and the single-factor obstruction

Since $\psi_3 = 3x(t + 28)$, substitution of (1.1) gives

$$R_3(t) = \frac{\beta}{81} \frac{f(t)}{t^3}, \qquad f(t) = t^3 - 672t^2 + 2352t + 21952. \qquad (4.1)$$

To verify (4.1), substitute (1.1) into $3x(t + 28)$ and divide by $3x(t + 28)$; the numerator factorization is $(t + 28)^3 - 756t^2 = f(t)$, and the remaining factors give the displayed constant and t^3 denominator. By (1.2),

$$A(P) := \chi_3(R_3(P)) = \chi_3(f(t(P))). \qquad (4.2)$$

Proposition 4.1. No function $g = c\psi_m$, with $c \in \mathbf{F}_p^*$, $p \nmid m$, and $n \nmid m$, satisfies the exact extractor equation (2.1).

Proof. First impose the symmetries of the target. Since M is even, $C(-P) = C(P)$. The identities $\psi_m(-P) = (-1)^{m-1}\psi_m(P)$ and $\eta(-1) = -1$ force m to be odd. Also $C(\phi P) = \beta C(P)$. The standard weight transformation of division polynomials is $\psi_m(\phi P) = \beta^{2(m^2-1)}\psi_m(P)$; applying η forces $3 \mid m$. Explicitly, $(p-1)/6 \equiv 1 \pmod 3$, so $\eta(\beta) = \beta$; the exponent $2(m^2-1)$ is zero modulo three when $3 \nmid m$ and one when $3 \mid m$. Thus $m = 3b$ with $m \equiv 3 \pmod 6$.

Equation (3.3) implies

$$\chi_3(R_{3b}(P)) = \chi_3(R_3([b]P)), \qquad (4.3)$$

because the remaining exponent is nine. Multiplication by b permutes the nonidentity rational group. Therefore (2.2) would force A to be constant there, with value $\xi^2 = \beta$.

The divisor of R_3 has eighteen simple zeros and poles of order nine at τ and $-\tau$. Its cubic Kummer sheaf is geometrically nontrivial, ramified at exactly eighteen geometric points. The tame character-sum bound on a genus-one curve gives

$$\left| \sum_{P \in E(\mathbf{F}_p)} \chi_3(R_3(P)) \right| \le 18\sqrt{p}. \qquad (4.4)$$

Here the function is interpreted by its finite nonzero value at O; all geometric branch points are nonrational. If the $n-1$ nonidentity terms were constant, the full sum would have magnitude at least $n-2$. But $n - 2 > 18\sqrt{p}$. This contradiction proves the proposition. □

Lemma 4.2 (the character-sum input). Let R define a finite-order rank-one Kummer local system on a smooth genus-one curve over $\mathbf{F}_p$, with exactly $s > 0$ geometric ramification points, all tame and with nontrivial inertia. Give its unramified extension the usual Frobenius trace at every remaining rational point. Then the sum of its traces has absolute value at most $s\sqrt{p}$.

Proof. Remove precisely the ramification points and denote the resulting open curve by U. Nontrivial geometric ramification excludes a constant constituent, so $H^0_c(U, \mathcal{L}) = H^2_c(U, \mathcal{L}) = 0$. The tame Euler-characteristic formula gives $\chi_c(U, \mathcal{L}) = 2 - 2g - s = -s$, hence $\dim H^1_c = s$. The finite-order local system is pure of weight zero; the curve trace formula and the Weil weight bound give eigenvalues of absolute value at most $\sqrt{p}$ in H^1_c [2,3]. As inertia has no invariants at the removed points, its middle-extension traces there are zero. Taking the trace proves the assertion. At a zero or pole whose valuation is divisible by the character order, use the unramified local extension, not an artificial zero value. □

For (4.4), all rational points, including O, are unramified and R_3 is a local unit there. Removing O changes a sum by at most one. This accounts explicitly for the constants $n-2$ and $+1$ used later.

The result closes this separable single-factor family even for enormous m. It does not close every product of factors or every compact arithmetic circuit.

5. Products and effective support

Consider

$$g = c \prod_{i=1}^{s} \psi_{m_i}^{e_i},$$

where $p \nmid m_i$, $n \nmid m_i$, and exponents may be reduced modulo six for the character question. The same symmetry tests give the necessary congruences

$$\sum_{2|m_i} e_i \equiv 0 \pmod 2, \qquad \sum_{3|m_i} e_i \equiv 1 \pmod 3. \tag{5.1}$$

Define $s_6(R_g)$ as the number of geometric points where the valuation of R_g is not divisible by six. This is the effective branch support for its sextic Kummer character, rather than its unreduced polynomial degree.

Proposed Theorem 5.1 (support obstruction). Any exact product extractor satisfying (2.1) must satisfy

$$n - 2 \le s_6(R_g)\sqrt{p}, \qquad s_6(R_g) \le 2 \sum_{e_i \not\equiv 0 \,(6)} (m_i^2 + 1). \tag{5.2}$$

Proof. The second bound follows by taking the union of the two degree-m_i^2 pullback supports and the two possible poles in (3.2); cancellations only decrease the support. For the first bound, a geometrically nontrivial rank-one Kummer sheaf gives the same genus-one estimate as in Section 4, with s_6 in place of eighteen.

One must justify nontriviality instead of assuming it. Suppose every valuation of R_g were divisible by six. The finite divisor $D = \operatorname{div}(g)$, taken modulo six, would satisfy $\Psi^* D = D$. Since Ψ is etale of degree three, the support of the pullback has three times as many geometric points as that of D. Thus this finite support must be empty. Write $\operatorname{div}(g) = 6D_0$ with D_0 rational. Its class is rational six-torsion in $\operatorname{Pic}^0(E) = E$, but the rational group has prime order coprime to six. Consequently D_0 is principal. A function with this rational divisor can be chosen over $\mathbf{F}_p$: its one-dimensional space of sections is Frobenius-stable, and a scalar rescaling removes the Frobenius multiplier (equivalently, use Hilbert 90 over a finite extension). Thus $g = c_0 g_0^6$ with $c_0 \in \mathbf{F}_p^*$. Its sextic character is constant, contradicting (2.1). Hence $s_6 > 0$ for an exact extractor, and the Kummer estimate applies. Covariance makes all nonidentity summands equal; adding the identity term gives the lower bound $n - 2$. □

The bound is on effective divisor complexity. Large indices can have short evaluation recurrences, so (5.2) is not a general running-time lower bound.

6. A cubic splitting field and geometric independence

The polynomial in (4.1) has

$$\operatorname{disc}(f) = 5334336^2 = 2^{12}3^{10}7^6, \qquad f(-7) = -27783. \tag{6.1}$$

To prove irreducibility explicitly, let $K = \mathbf{F}_{p^3}$ and choose ρ with $\rho^3 = 19 + \beta$. The exact

residue identity $(19+\beta)^{(p-1)/3} = \beta$ gives $\rho^p = \beta\rho$. One root of f is

$$\alpha = 224 + 84\left(\rho + \frac{7}{\rho}\right), \qquad f(t) = N_{K/\mathbf{F}_p}(t-\alpha). \tag{6.2}$$

Verification of the root and its degree. The identity $(19+\beta)(19+\beta^2) = 343$ gives $\rho^3 + (7/\rho)^3 = 37$. Thus $u = \rho + 7/\rho$ satisfies $u^3 - 21u - 37 = 0$, and direct substitution yields

$$f(224 + 84u) = 84^3(u^3 - 21u - 37).$$

The residue check in Appendix A makes $19+\beta$ a noncube, so $1, \rho, \rho^2$ is a basis of K over $\mathbf{F}_p$. If $\rho + 7/\rho$ were Frobenius-fixed, then $(\beta-1)\rho + 7(\beta^2-1)/\rho = 0$, forcing $\rho^2 \in \mathbf{F}_p$, a contradiction. Hence α has degree three. Its three Frobenius conjugates are exactly the roots of the monic cubic f, proving both irreducibility and (6.2).

Consequently, with $\tilde{\chi}_3(z) = z^{(p^3-1)/3}$,

$$A(P) = \tilde{\chi}_3(t(P) - \alpha). \tag{6.3}$$

This is a linear Kummer symbol over a small extension. It is not itself the usual three-isogeny descent function. Over a field containing $s = \sqrt{7}$, that descent-type function is $y - s$, with divisor $3(\tau) - 3(O)$. In contrast, $t - \alpha$ has six simple zeros.

Proposition 6.1. The smooth normalization of

$$\mathcal{C}: \quad y^2 = x^3 + 7, \qquad z^3 = x^3 - \alpha$$

has genus seven. Its pullback by $[b]$, for $b > 0$ and $p \nmid b$, has genus $6b^2 + 1$.

Proof. The cubic Kummer cover of E has six simple branch points and an unramified pole of order six at O. Riemann–Hurwitz gives $2g - 2 = 6(3-1) = 12$. An etale pullback by $[b]$ gives $6b^2$ branch points, yielding $g = 6b^2 + 1$. □

Proposition 6.2 (positive-index independence). For distinct positive integers $b_1, \ldots, b_s$, all prime to p, the classes of

$$h_b(P) = t([b]P) - \alpha$$

are linearly independent in $\bar{K}(E)^*/\bar{K}(E)^{*3}$. Their cube roots define a cover with group $(\mathbf{Z}/3)^s$.

Proof. The six zeros of $t - \alpha$ are points of exact order nine: they belong to $\ker \Psi^3 \setminus \ker \Psi^2$. For the largest index b, choose a preimage Q of one such zero under $[b]$ with exact order $9b$. Such a preimage exists by choosing compatible bases of the prime-to-p Tate modules: a primitive order-nine vector lifts under $[b]$ to an order-$9b$ vector, and the prime-to-three factors of b can be supplied in the other torsion components. If $[c]Q$ were a zero for a smaller positive index c, its order would be nine, forcing $\gcd(c, 9b) = b$ and hence $b \mid c$, impossible. Nor is it a

pole for a smaller index. The valuation at Q isolates the exponent of h_b in any proposed cube relation. It must be zero modulo three. Descending through the indices proves independence. □

The restriction to positive indices is explicit: $h_b = h_{-b}$, so unrestricted "distinct indices" would be false. Also, functions with distinct geometric indices may agree on the finite rational group when the indices coincide modulo n. Geometric Kummer independence must never be confused with independence of finite samples or with the absence of every finite-orbit character identity.

7. The ternary sequence and its holonomy

Set

$$P_r = \Psi^r G = [h^r]G, \quad t_r = F^r(t(G)), \quad A(P_r) = \beta^{a_r}, \qquad a_r \in \mathbf{F}_3.$$

Since A is sixfold invariant, (a_r) is M-periodic. If $b = h^s \pmod n$, then $A([b]P_r) = A(P_{r+s})$. Thus scalar pullbacks of this atom are shifts of a single cyclic sequence, and a multiplicative character relation becomes an affine relation among its shifts.

Define $\chi_3(\psi_3(P_r)) = \beta^{c_r}$. The ratio defining R_3 gives

$$a_r = c_{r+1} - c_r. \qquad (7.1)$$

The c_r are not M-periodic. Since $h^M = -\lambda^2$ and $\psi_3(\phi P) = \beta\psi_3(P)$, the orientation identities yield

$$c_{r+M} = c_r + 1, \qquad \sum_{r=0}^{M-1} a_r = 1 \pmod 3. \qquad (7.2)$$

For the sign and exponent in (7.2), $P_{r+M} = -\phi^2 P_r$. The polynomial ψ_3 is even under negation and has ϕ-weight β, so $\psi_3(P_{r+M}) = \beta^2\psi_3(P_r)$. Since $(p-1)/3 \equiv 2 \pmod 3$, applying χ_3 multiplies by $\beta^4 = \beta$. This gives the increment one. Summing (7.1) over a full quotient cycle telescopes to that increment, rather than to zero.

With $d_r = c_r - r$ and $b_r = a_r - 1$, one obtains genuinely periodic sequences satisfying

$$b_r = d_{r+1} - d_r, \qquad \sum_r b_r = 0. \qquad (7.3)$$

This is nontrivial cyclic holonomy, not a globally periodic primitive c_r.

For clarity, fix the backward cyclic shift $(Sb)_r = b_{r-1}$ and write $B(X) = \sum_r b_r X^r$ in $\mathbf{F}_3[X]/(X^M - 1)$. Its minimal annihilator is

$$m_b(X) = \frac{X^M - 1}{\gcd(B(X), X^M - 1)}. \qquad (7.4)$$

Indeed, $E(S)b = 0$ is equivalent to $(X^M - 1) \mid E(X)B(X)$. A forward-shift convention replaces the annihilator by its reciprocal; the convention is therefore part of the statement. Since $3 \nmid M$, $X^M - 1$ is square-free, and $B(1) = 0$.

Open problem. Does $\gcd(B(X), X^M - 1) = X - 1$ hold? This maximal-complexity possibility is not proved here. The local results below do not settle that global question.

8. Disjoint branch layers and recurrence bounds

Let $K_j = \ker \Psi^j$ over an algebraic closure, so $|K_j| = 3^j$. The cubic Kummer branch support of R_3 is

$$B_0 = K_3 \setminus K_2.$$

For $r_i(P) = R_3(\Psi^i P)$, the effective cubic support is

$$B_i = K_{i+3} \setminus K_{i+2}, \qquad |B_i| = 18 \cdot 3^i. \tag{8.1}$$

To justify (8.1), observe that $K_2 = E[3]$ by $\Psi^2 = -[3]\phi^2$. In (3.2) with $m = 3$, the eighteen preimages of τ and $-\tau$ are precisely $K_3 \setminus K_2$, with multiplicity one; the only poles are $\tau, -\tau$, of order nine. Pullback by the etale map Ψ^i gives $\Psi^{-i}(K_j) = K_{i+j}$ and preserves the local multiplicities. Every pulled-back pole still has valuation divisible by three.

These supports are pairwise disjoint. The poles have valuations divisible by three and do not enter the cubic conductor. Therefore a nonzero exponent vector $e_i \in \{0, 1, 2\}$ gives exactly

$$s\left(\prod_i r_i^{e_i}\right) = 18 \sum_{e_i \neq 0} 3^i. \tag{8.2}$$

There is no cancellation between different levels modulo cubes.

Proposed Theorem 8.1 (short affine recurrences). A nonzero relation

$$\sum_{i=0}^{k-1} e_i a_{r+i} = c \qquad \text{for every } r$$

requires

$$n - 2 \leq 18 \left(\sum_{e_i \neq 0} 3^i\right) \sqrt{p} \leq 9(3^k - 1)\sqrt{p}. \tag{8.3}$$

Proof. The relation makes the associated nontrivial cubic character product constant on every nonidentity rational point, because the primitive orbit covers them all. Its full rational-point sum has magnitude at least $n - 2$, while the Kummer bound and (8.2) give the upper bound. □

The exact integer comparisons printed in Appendix A show that $9(3^{78}-1)\sqrt{p} < n-2$, whereas the corresponding sufficient inequality at $k = 79$ no longer holds. Thus the theorem excludes every nonzero affine recurrence supported within **78 consecutive positions**, indexed $0, \dots, 77$. It does not assert existence of a recurrence at length 79.

9. Every short word occurs

For $w \in \mathbf{F}_3^k$, let N_w count nonidentity rational starting points for which the next k labels are w. Character orthogonality and (8.2) give

$$\left| N_w - \frac{n-1}{3^k} \right| \le 6(3^k-1)\sqrt{p} + 1 - 3^{-k}. \qquad (9.1)$$

Proof. Expand the indicator of each coordinate equality using the three additive characters of $\mathbf{F}_3$. The zero exponent vector contributes $(n-1)/3^k$. Each nonzero vector contributes a character sum bounded by $s(e)\sqrt{p}+1$, the additional one accounting for removing O. In the sum over all exponent vectors, the ith coordinate is nonzero in $2 \cdot 3^{k-1}$ cases. Hence

$$\frac{1}{3^k}\sum_e s(e) = \frac{18}{3^k}(2 \cdot 3^{k-1}) \sum_{i=0}^{k-1} 3^i = 6(3^k-1).$$

The $3^k - 1$ nonzero vectors contribute the remaining $1 - 3^{-k}$. □

Each quotient starting state corresponds to six rational points with the same word, so the number of occurrences on the quotient cycle is $N_w/6$. A sufficient condition for positivity is

$$n - 3^k > 6 \cdot 3^k(3^k-1)\sqrt{p}. \qquad (9.2)$$

It holds at $k = 39$ and therefore for every smaller positive length. At $k = 40$ this estimate is insufficient. Accordingly **every ternary word of length at most 39 occurs**. This is a consequence of the conductor calculation, not an extrapolation from a finite simulated prefix.

A deterministic autonomous symbol generator reproducing the sequence would need at least 3^{39} distinguishable starting states, since two states producing different length-39 words cannot be the same state. This is a state-count consequence, not a time lower bound: roughly 62 bits already suffice to name that many states. It does not exclude a compact description with a large state space.

10. Kummer towers and the distinction between driven and autonomous states

The disjoint supports show that adjoining cube roots of $r_0, \dots, r_{k-1}$ gives geometric deck group $(\mathbf{Z}/3)^k$. Pullback by Ψ shifts the corresponding Kummer classes. Thus each extra level adds genuinely new geometric branch data.

A related rational-function formulation uses $u_j(t) = F^j(t) + 28$. Its new zero set is $F^{-j}(-28)$, consisting of 3^j distinct points. The postcritical values of F are $0, \infty, -7$, as follows from $F'/F = (t-56)/(t(t+28))$. Thus these inverse images are unramified and the zero sets are

disjoint: a shared point would make -28 periodic, whereas $-28 \mapsto 0 \mapsto \infty \mapsto \infty$. At a zero of u_j, a later u_{j+2} has a pole of order six, and all subsequent poles there have that same order; they vanish modulo three. Alternatively, the latest nonzero exponent in any finite relation is isolated at a fresh simple zero. This proves that the classes $[u_j]$ are independent modulo cubes. In particular, there is no fixed finite-dimensional pullback-stable Kummer span containing $[t+28]$. This conclusion is restricted to this generic Kummer representation; it is not a theorem against all finite-field algorithms or all automata.

There is also a three-valued extension driven by the quotient. Let $D(P) = \chi_3(x(P))$. Because $\chi_3(-\beta/3) = 1$, the explicit x map gives

$$\frac{D(\Psi P)}{D(P)} = \chi_3(t(P) + 28), \qquad A(P_r) = \frac{D(P_{r+2})}{D(P_{r+1})}. \tag{10.1}$$

This is a small **fiber state driven by the quotient coordinate**. The base coordinate still ranges over M states. It is not an autonomous three-state generator of the ternary sequence, and does not contradict the block bound.

11. Explicit higher-genus quotients and elliptic factors

The curve $\mathcal{C}$ of Proposition 6.1 has a visible subgroup $A \simeq C_3 \times C_3$ generated by

$$a : (x, z, y) \mapsto (\beta x, z, y), \qquad b : (x, z, y) \mapsto (x, \beta z, y).$$

Negation of y gives a commuting involution. No assertion that this subgroup is the entire automorphism group is needed.

Proposition 11.1 (four quotients). Put $R = (\alpha + 7)/7$. The four order-three subgroups of A give quotient curves of genera $1, 1, 2, 3$, respectively. They have the following models, interpreted through smooth projective normalization:

$$E : y^2 = x^3 + 7, \qquad E_\alpha : y^2 = z^3 + \alpha + 7,$$

$$D_2 : Y^2 = 7(w^3 - 1)(Rw^3 - 1),$$

$$D_3 : v^3 = (y^2 - 7)(y^2 - \alpha - 7).$$

Moreover, over $K = \mathbf{F}_{p^3}$,

$$\mathrm{Jac}(\mathcal{C}) \ \sim_K \ E \times E_\alpha \times \mathrm{Jac}(D_2) \times \mathrm{Jac}(D_3). \tag{11.1}$$

Proof. The quotients by $\langle b \rangle$ and $\langle a \rangle$ forget z and x, giving the two elliptic curves. For $\langle ab \rangle$, the invariant $w = x/z$ satisfies

$$w^3 = \frac{y^2 - 7}{y^2 - \alpha - 7}.$$

Setting $Y = y(w^3 - 1)$ gives D_2. Its degree-six polynomial has six distinct roots because $R \neq 0, 1$, so its genus is two. For $\langle ab^{-1} \rangle$, the invariant $v = xz$ gives D_3. The degree-four polynomial in y has four distinct roots, and infinity is also ramified in its cubic cover of $\mathbf{P}^1_y$. Riemann–Hurwitz gives $2g - 2 = 3(-2) + 5 \cdot 2 = 4$, hence genus three. The conditions used here follow from $f(0) \neq 0$ and $f(-7) \neq 0$.

For the isogeny, use the sum of pullback maps from the four quotient Jacobians. On first cohomology with characteristic-zero coefficients, every nontrivial character of A has a unique order-three kernel. Its eigenspace therefore occurs in exactly one of the four quotient invariant spaces. The trivial-character part is zero because $\mathcal{C}/A = \mathbf{P}^1_y$. These invariant spaces consequently form a direct sum equal to $H^1(\mathcal{C})$; their dimensions are $2, 2, 4, 6$. The pullback map from the product has full rank on Tate modules, so is an isogeny. All quotient maps are defined over K, giving (11.1) over that field. The functorial pullback and norm maps on Jacobians used here are standard [6]. □

Proposition 11.2 (reciprocal form and its two elliptic quotients). One has $N_{K/\mathbf{F}_p}(R) = 81$. The element R is a square and a noncube in K. Choose $\gamma \in K$ with $\gamma^2 = R$, and put $S = \gamma + \gamma^{-1}$. Over the cubic extension $L = \mathbf{F}_{p^9}$, D_2 is isomorphic to

$$D_0 : Y^2 = 7(X^6 - SX^3 + 1). \qquad (11.2)$$

The curve D_0, itself defined over K, has degree-two maps to the elliptic curves

$$E_+ : V_+^2 = 7(u + 2)(u^3 - 3u - S),$$

$$E_- : V_-^2 = 7(u - 2)(u^3 - 3u - S). \qquad (11.3)$$

Proof. The norm equals $-f(-7)/7^3 = 81$. Compatibility of norm with residue characters gives $\eta_{6,K}(R) = \eta_{6,p}(81) = \beta$: indeed $\chi_3(81) = \chi_3(\beta) = \beta^2$ by (1.2), and 81 is a square, fixing the sextic value. Thus R is a square but not a cube. Choose a_0 with $a_0^3 = \gamma^{-1}$. It lies in L, and substituting $w = a_0 X$ gives (11.2).

The involutions $\sigma(X, Y) = (X^{-1}, Y/X^3)$ and $-\sigma$, where the minus is the hyperelliptic involution, have invariant functions

$$u = X + X^{-1}, \qquad V_\pm = Y(X \pm 1)/X^2.$$

Substitution gives (11.3). Both quartics are nonsingular because $S^2 \neq 4$, and each has the rational point $u = \mp 2, V_\pm = 0$, so their smooth models are elliptic curves. The degree-two quotient maps will be denoted $\pi_\pm$. □

Proposition 11.3 (a degree-three isogeny). The elliptic curves E_+ and E_- in (11.3)

are three-isogenous over K. Also $\mathrm{Jac}(D_0) \sim_K E_+ \times E_-$; consequently this further splitting applies to $\mathrm{Jac}(D_2)$ over L.

Proof. Let $J = \mathrm{Jac}(D_0)$ and let r be the automorphism induced by $(X, Y) \mapsto (\beta X, Y)$. The quotient by r has the conic model $Y^2 = 7(Z^2 - SZ + 1)$, so $1 + r + r^2 = 0$ on J. Also $\sigma r \sigma = r^{-1}$. The pullback and norm identities for the two degree-two maps are

$$\pi_+^* \pi_{+*} = 1 + \sigma, \quad \pi_-^* \pi_{-*} = 1 - \sigma, \quad \pi_{\pm *} \pi_\pm^* = [2].$$

The first two eigenspaces are complementary on rational Tate modules, giving the claimed product isogeny. Define

$$\varphi = \pi_{-*}\, r\, \pi_+^* : E_+ \longrightarrow E_-.$$

Under the principal polarizations its dual is $\widehat{\varphi} = \pi_{+*} r^{-1} \pi_-^*$. Let $A = \pi_{+*} r \pi_+^*$. Since σ acts trivially on $\pi_+^* E_+$ and $\pi_{+*}\sigma = \pi_{+*}$, conjugating by σ gives $A = \pi_{+*} r^{-1} \pi_+^*$. Therefore $2A = \pi_{+*}(r + r^{-1})\pi_+^* = [-2]$, and $A = [-1]$ because an elliptic-curve endomorphism group is torsion-free. Now

$$\widehat{\varphi}\varphi = \pi_{+*} r^{-1}(1 - \sigma) r \pi_+^* = \pi_{+*}(1 - r\sigma)\pi_+^* = [2] - A = [3].$$

Thus φ is nonzero and has degree three. Every map in this construction is defined over K, and characteristic is not three. □

The field distinction matters. The reciprocal curve D_0 and the isogeny in Proposition 11.3 are defined over K, but the change $D_2 \simeq D_0$ uses $a_0 \in L$. The assertion that the original factor $\mathrm{Jac}(D_2)$ splits over K does not follow; its twist and Frobenius descent require a separate argument. Likewise, a full classification of the inverse-iteration tower or an unrestricted cyclic-annihilator theorem is not claimed here.

12. Conclusion

The exact sextic target leads to a covariance equation whose nontrivial character sums obstruct a single division polynomial and all products of sufficiently small effective support. Along the primitive quotient orbit, the cubic atom becomes a cyclic sequence with explicitly computed holonomy, disjoint ramification layers, no affine recurrence in 78 consecutive positions, and occurrence of every ternary word of length at most 39. Appendix A supplies the complete finite arithmetic underlying these numerical thresholds.

The higher-genus constructions provide explicit quotient curves and isogenies, not an unknown-scalar multiplication operation. Geometric independence does not imply statistical independence on a finite orbit. The full cyclic annihilator and the descent of the further genus-two splitting remain separate questions. No faster private-key-recovery algorithm follows from the statements proved here.

Appendix A. Complete arithmetic certificate

This appendix gives all data needed to certify the scalar order; it does not rely on an external computation file. The prime coordinate modulus and the curve parameters are the published SEC 2 parameters. The order n, the three large factors of $n-1$, and every auxiliary prime used below are certified by the following elementary criterion.

Lucas criterion. Suppose the complete factorization $q-1=\prod t^{e_t}$ is known, every t has already been proved prime, and an integer a satisfies

$$a^{q-1}\equiv 1 \pmod{q}, \qquad \gcd(a^{(q-1)/t}-1,q)=1 \quad \text{for every } t \mid q-1.$$

Then q is prime and a has order $q-1$ modulo q. Indeed, for every prime divisor ℓ of q, the displayed conditions force the order of a modulo ℓ to contain the full power of every prime dividing $q-1$. Thus $q-1 \mid \ell-1$, so $\ell \geq q$ and $\ell = q$.

Use the parameters n, λ of Section 1 and $h=\lambda+2$. The labels in the table mean

$$q_1 = 107361793816595537, \qquad q_2 = 174723607534414371449,$$

$$q_3 = 341948486974166000522343609283189,$$

$$r_0 = 290476118734425756474977581 79.$$

Starting with the prime 2, the rows are in increasing numerical order. Each row states a complete factorization and a Lucas witness. Integer multiplication, modular exponentiation, and the Euclidean algorithm verify every row. No probabilistic primality test is required.

q	Factorization of $q-1$	Witness a
3	2	2
5	2^2	2
7	$2\cdot 3$	3
11	$2\cdot 5$	2
13	$2^2\cdot 3$	2
17	2^4	3
19	$2\cdot 3^2$	2
23	$2\cdot 11$	5
29	$2^2\cdot 7$	2
37	$2^2\cdot 3^2$	2
41	$2^3\cdot 5$	6
59	$2\cdot 29$	2
67	$2\cdot 3\cdot 11$	2
73	$2^3\cdot 3^2$	5
97	$2^5\cdot 3$	5
109	$2^2\cdot 3^3$	6

q	Factorization of $q-1$	Witness a
113	$2^4 \cdot 7$	3
149	$2^2 \cdot 37$	2
199	$2 \cdot 3^2 \cdot 11$	3
293	$2^2 \cdot 73$	2
461	$2^2 \cdot 5 \cdot 23$	2
631	$2 \cdot 3^2 \cdot 5 \cdot 7$	3
797	$2^2 \cdot 199$	2
1409	$2^7 \cdot 11$	3
1871	$2 \cdot 5 \cdot 11 \cdot 17$	14
2011	$2 \cdot 3 \cdot 5 \cdot 67$	3
2731	$2 \cdot 3 \cdot 5 \cdot 7 \cdot 13$	3
2861	$2^2 \cdot 5 \cdot 11 \cdot 13$	2
4051	$2 \cdot 3^4 \cdot 5^2$	10
9349	$2^2 \cdot 3 \cdot 19 \cdot 41$	2
16699	$2 \cdot 3 \cdot 11^2 \cdot 23$	3
28181	$2^2 \cdot 5 \cdot 1409$	2
85831	$2 \cdot 3 \cdot 5 \cdot 2861$	3
120233	$2^3 \cdot 7 \cdot 19 \cdot 113$	3
305873	$2^4 \cdot 7 \cdot 2731$	3
1627771	$2 \cdot 3 \cdot 5 \cdot 29 \cdot 1871$	3
4681609	$2^3 \cdot 3 \cdot 97 \cdot 2011$	23
44706919	$2 \cdot 3 \cdot 797 \cdot 9349$	6
545358713	$2^3 \cdot 41 \cdot 59 \cdot 28181$	5
297159362677	$2^2 \cdot 3^2 \cdot 11 \cdot 461 \cdot 1627771$	2
q_1	$2^4 \cdot 16699 \cdot 85831 \cdot 4681609$	3
q_2	$2^3 \cdot 17 \cdot 59 \cdot 4051 \cdot 120233 \cdot 44706919$	3
r_0	$2 \cdot 293 \cdot 305873 \cdot 545358713 \cdot 297159362677$	2
q_3	$2^2 \cdot 3^3 \cdot 109 \cdot r_0$	2
n	$2^6 \cdot 3 \cdot 149 \cdot 631 \cdot q_1 \cdot q_2 \cdot q_3$	h

The last row uses h itself. It simultaneously proves that n is prime and that h has order $n-1$. Its seven proper-order tests are precisely those for the distinct factors $2, 3, 149, 631, q_1, q_2, q_3$; in particular, the certificate is not a traversal of the curve group.

The following additional checks use the same stated parameters. They can be reproduced by the short exact-arithmetic code printed below; exponentiation is repeated squaring. Here hexadecimal strings are concatenated before conversion.

```
from math import gcd
p = 2**256 - 2**32 - 977
n = int('FFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFE'
        'BAAEDCE6AF48A03BBFD25E8CD0364141', 16)
beta = int('7ae96a2b657c07106e64479eac3434e9'
           '9cf0497512f58995c1396c28719501ee', 16)
lam = int('5363ad4cc05c30e0a5261c028812645a'
```

```
          '122e22ea20816678df02967c1b23bd72', 16)
h, M = lam + 2, (n - 1) // 6
factors = [2, 3, 149, 631, 107361793816595537,
           174723607534414371449,
           341948486974166000522343609283189]
assert (beta*beta + beta + 1) % p == 0
assert (lam*lam + lam + 1) % n == 0
assert pow(h, n-1, n) == 1
for t in factors:
    assert gcd(pow(h, (n-1)//t, n)-1, n) == 1
assert pow(p, M, n) == 1
for t in factors:
    if M % t == 0:
        assert pow(p, M//t, n) != 1
assert pow(h, M, n) == (-lam*lam) % n
assert M % 3 == 1 and p % 9 == 7
assert pow((beta*pow(81, -1, p)) % p, (p-1)//3, p) == 1
assert pow(19+beta, (p-1)//3, p) == beta
assert pow((-beta*pow(3, -1, p)) % p, (p-1)//3, p) == 1
assert (n-2)**2 > (9*(3**78-1))**2*p
assert (n-2)**2 <= (9*(3**79-1))**2*p
assert (n-3**39)**2 > (6*3**39*(3**39-1))**2*p
assert (n-3**40)**2 <= (6*3**40*(3**40-1))**2*p
```

The code uses ordinary arbitrary-precision integers; negative modular exponents denote inverses. The four squared inequalities are used only with positive sides. They therefore certify, without floating-point rounding, the thresholds in the character-sum applications. These finite arithmetic certificates support the explicitly stated parameters and identities; they are not a substitute for the geometric proofs.

Research provenance, contributions, and verification status

Origin and development of the investigation

This mathematical investigation was initiated and directed by Louis Desbiens and developed through extensive interaction with OpenAI's ChatGPT. Louis Desbiens identified the original research questions, set the objectives, guided the investigation, assessed its progress at a conceptual level, and selected the material for public dissemination.

ChatGPT contributed substantially to mathematical exploration, including alternative formulations, derivation of algebraic identities, formulation and examination of conjectures and proposed proofs, identification of potential counterexamples and exceptional cases, design and execution of symbolic and numerical checks, organization of relevant literature, and drafting and revision of the text. These contributions went well beyond linguistic or editorial assistance.

Attribution and scientific authorship

Louis Desbiens is identified as the **human initiator and curator**, not as an independent mathematical authority certifying the results. OpenAI's ChatGPT is identified as the **AI mathematical research contributor**, not as an accountable conventional scholarly coauthor. These designations describe the provenance without implying that the human curator personally derived or independently verified all the arguments.

Mathematical verification status

The candidate mathematical contributions have not undergone independent specialist validation, formal proof verification, or an exhaustive originality and priority review. In particular, independent examination is needed of Kummer character-sum hypotheses and conductors; independence and branch-support arguments underlying the 78/39 thresholds; quotient genera, Jacobian splitting, isogeny and fields of definition. Locally displayed proofs are proposed arguments. Established results cited from public literature should be assessed separately from new deductions; neither an AI-generated proof nor a successful computational check is by itself an independently certified theorem. Individual statements may be correct, partly correct, incomplete, or incorrect.

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Computational evidence and source availability

Numerical evaluations helped develop the initial intuition. The public repository contains the available scripts, archived results, and a claim-by-claim verification coverage guide. These are scoped finite arithmetic, symbolic, and numerical checks, not independent verification of the general mathematical results. Historical exploratory calculations for which independent source data are unavailable are not treated as reproducible evidence.

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